

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	If a body A exerts a force on a body B, then B exerts an equal and opposite force on A (1) Accept: For every action there is an <u>equal and opposite</u> reaction	1			1		
		(ii)	1. <u>Gravitational</u> force of drone on Earth (1) (All required) 2. Force of rotors on air (1) (All required) Accept rotors pushing on air		2		2		
	(b)	(i)	The rate of change of momentum of an object is proportional to the resultant force acting on it, (and takes place in the direction of that force) [or equivalent] Acceptable minimum answers: Force is proportional (or equal) to the rate of change of momentum / force = <u>change</u> in momentum ÷ time Or in symbols e.g. $F = \frac{\Delta p}{\Delta t}$ and Δp defined	1			1		
		(ii)	Substitution into $4\pi r^2 \rho v^2$ i.e. $4 \times \pi \times (5 \times 10^{-2})^2 \times 1.3 \times (22)^2$ (1) F (= weight) = 19.7[7 N] or 19.8 [N] seen (1) Assume answer represents weight of drone, even if not stated. Answer gains both marks	1	1		2	2	